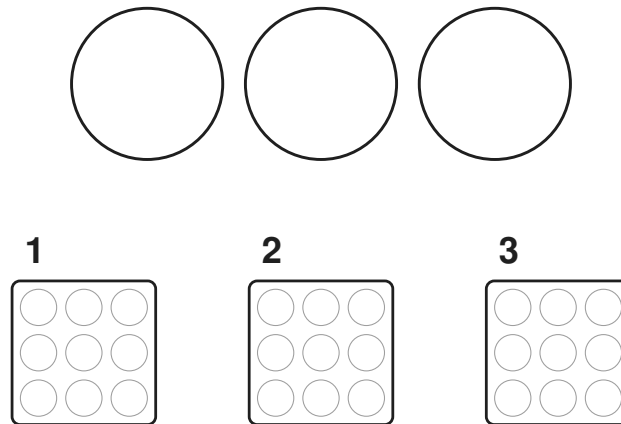
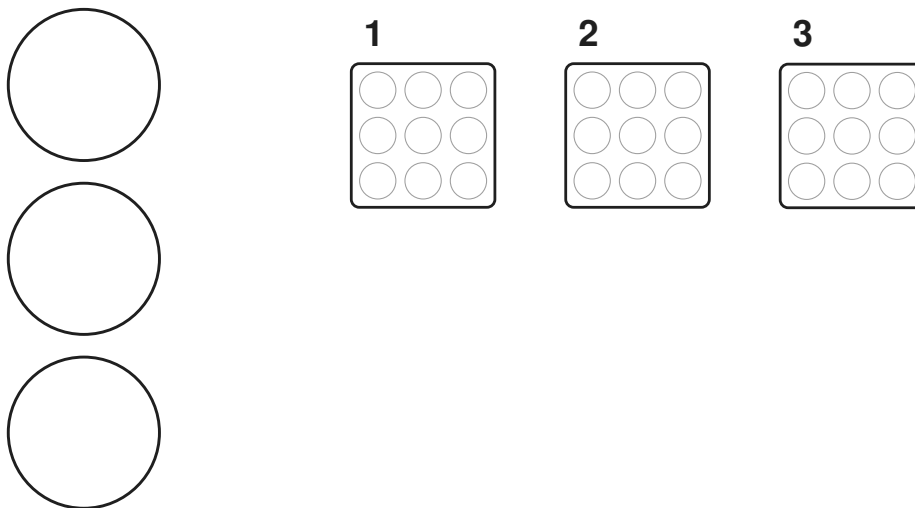


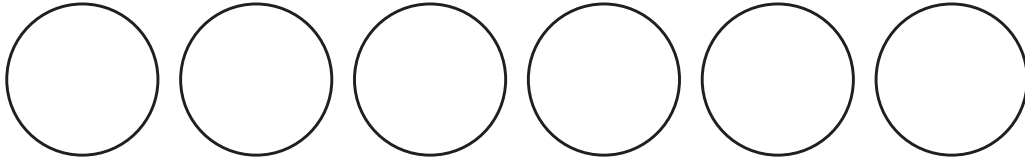
Problem 1: Put 3 counters on the 3 big circles. They make one row. Now make a move. A move: take one counter out of every row, even out of a row with only one counter. Put all the counters you took together as a new row. Put the longest row on top. Color the little circles in box 1 to show your rows. Make another move and color box 2. Make one more move and color box 3. Point to two boxes that look the same.



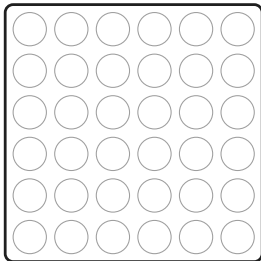
Problem 2: Put 3 counters on the big circles: three rows with one counter in each row. Make a move and color box 1. Make a move and color box 2. Make a move and color box 3. Find the box where the game stopped changing. Then find the same picture in Problem 1.



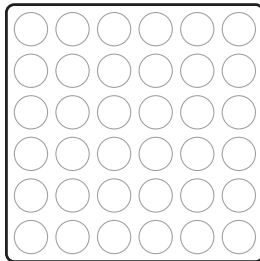
Problem 3: Put 6 counters on the 6 big circles. Make a move and color box 1. Keep making moves. Color the next box after each move. Stop when the picture stops changing. Then draw a line around the outside of the colored circles in your last box.



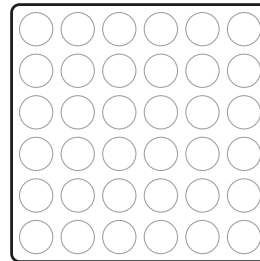
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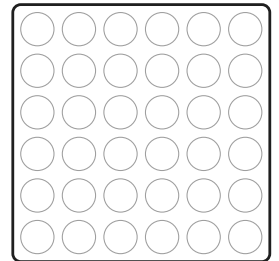
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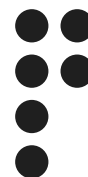
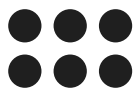
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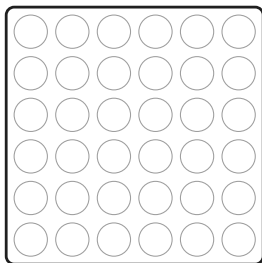
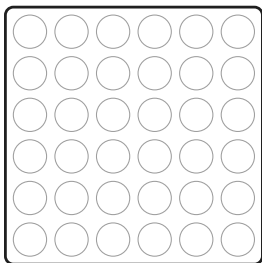
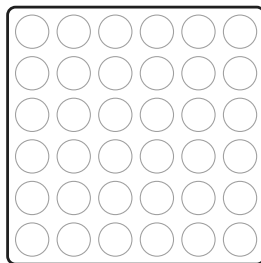
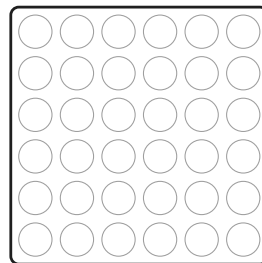
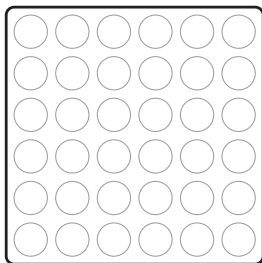
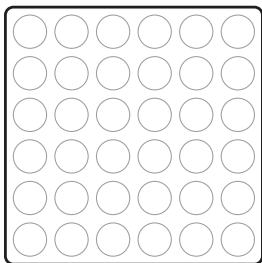
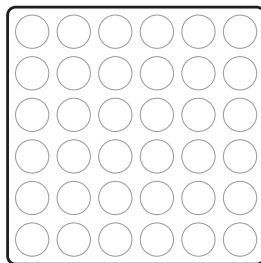
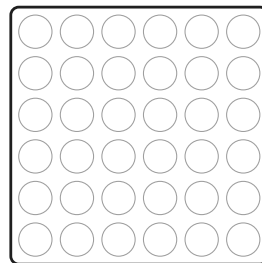
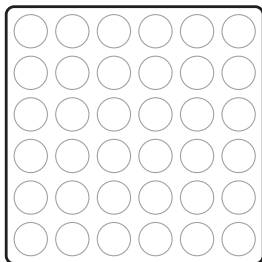
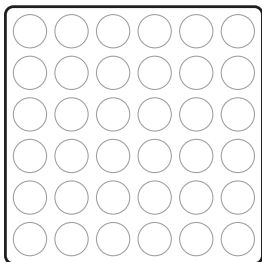
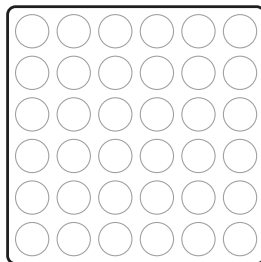
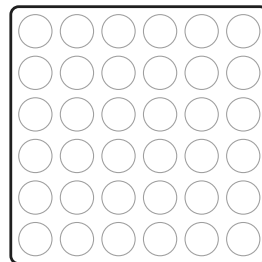
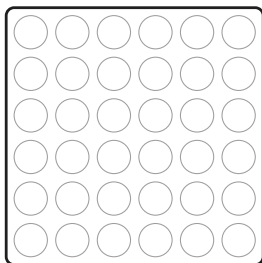
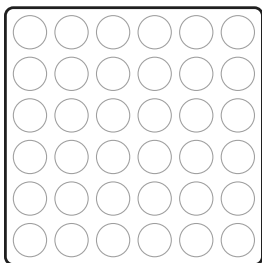
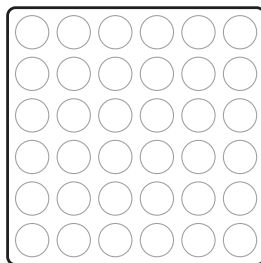
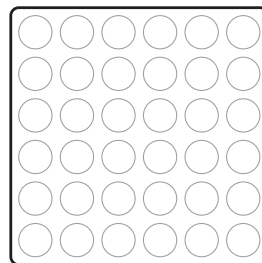


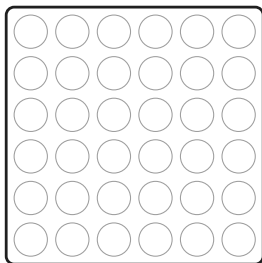
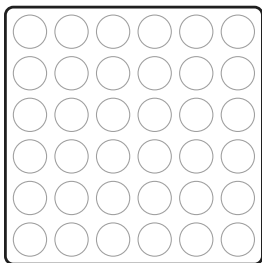
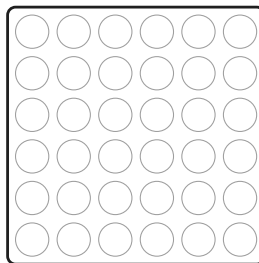
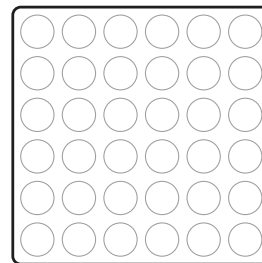
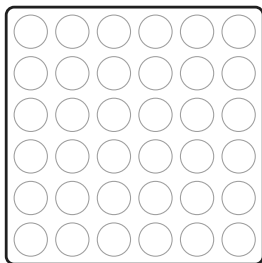
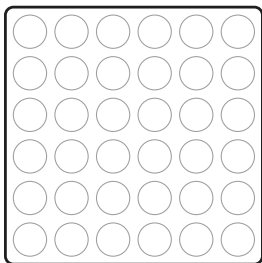
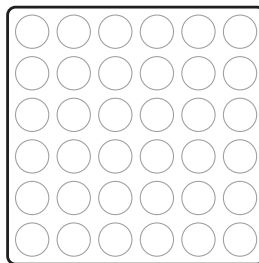
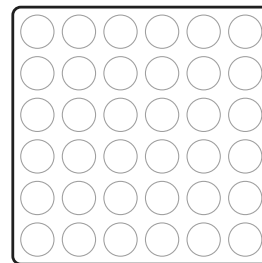
4



Problem 4: Choose a start with 6 counters: one of the five pictures below, or one you make up. Build it with counters. Color box 0 on the next page to show your start. Make a move and color box 1. Keep going. Stop when the picture stops changing. Then choose a different start and fill the boxes marked start B, and another start for the boxes marked start C.



Problem 4, start A**0****1****2****3****4****5****6****7****Problem 4, start B****0****1****2****3****4****5****6****7**

Problem 4, start C**0****1****2****3****4****5****6****7**

Problem 4, continued: Count the moves each start took before the picture stopped changing. Write the number in the box. Then draw the stopping picture from start A in the big box. Draw the stopping picture from start B next to it. Are they the same?

A

B

C

Problem 5: Put 4 counters in one row on the table. Make a move. Put the new picture on mat 1, rows with the longest on top. Now take 4 new counters and build a copy of mat 1 on the table. Make a move on the copy. Put the new picture on mat 2. Keep going the same way: copy the newest mat with new counters, make a move, put the new picture on the next mat. Stop when all 6 mats are full. Draw a line between two mats that show the same picture. Find every pair.

1

2

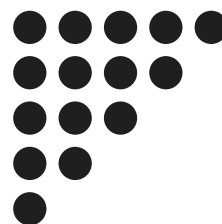
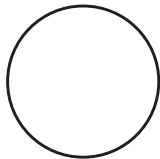
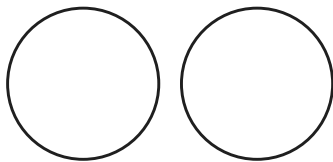
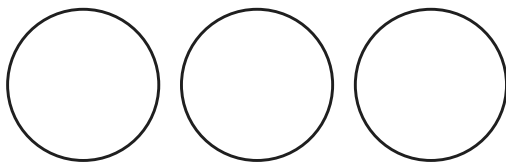
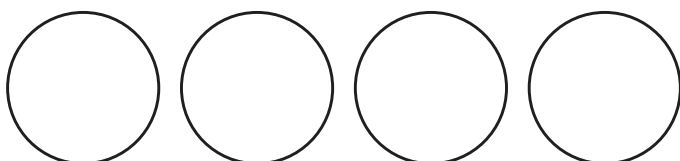
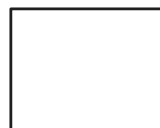
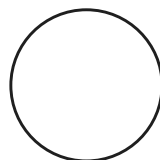
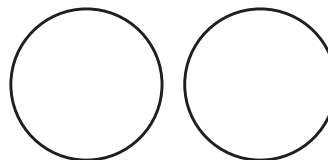
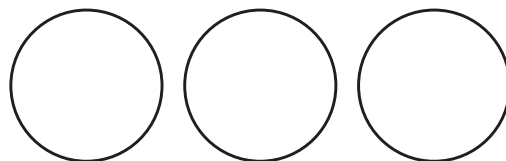
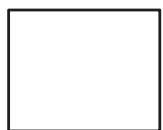
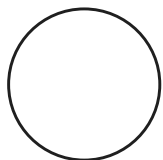
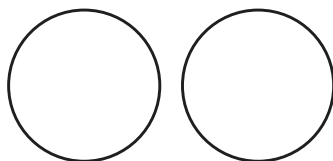
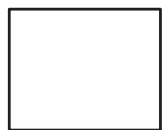
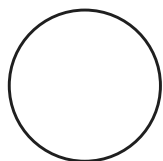
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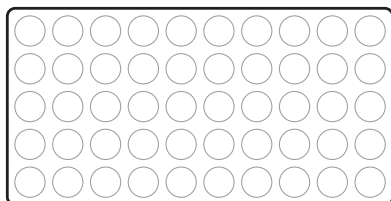
Problem 6: Cover every circle with a counter. Count the counters in each staircase. Write the number in the box under it. Then build the next staircase on the table. It has one more row. Count its counters and write the number in the last box. Now build the staircase with 4 rows on the table with 10 counters and make a move. Put the longest row on top. Is it the same picture as before the move?



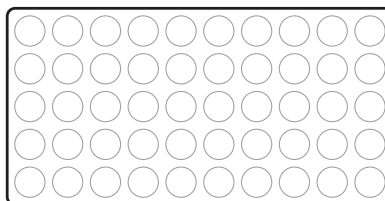
Problem 7: Put 10 counters in one row on the table, like the picture. Make a move and color box 1. Keep going. Color the next box after each move. Stop when the picture stops changing. Then, in every box, draw a line around all the rows except the top row.



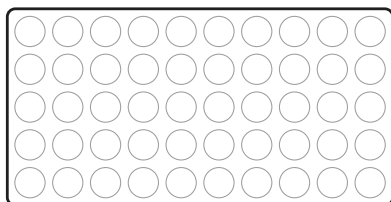
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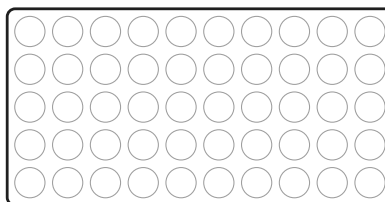
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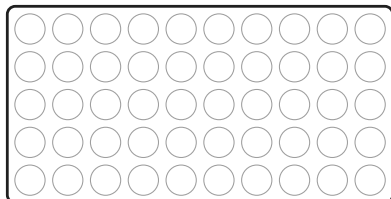
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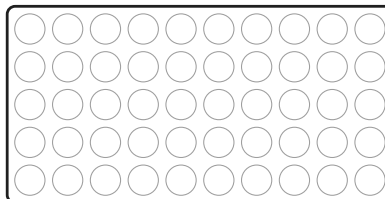
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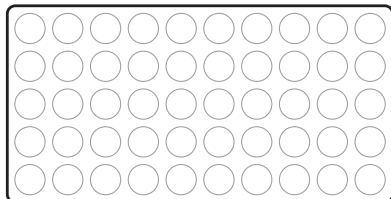
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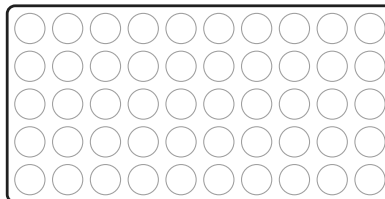
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7



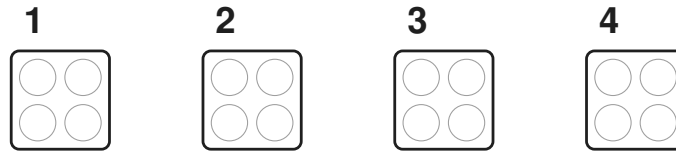
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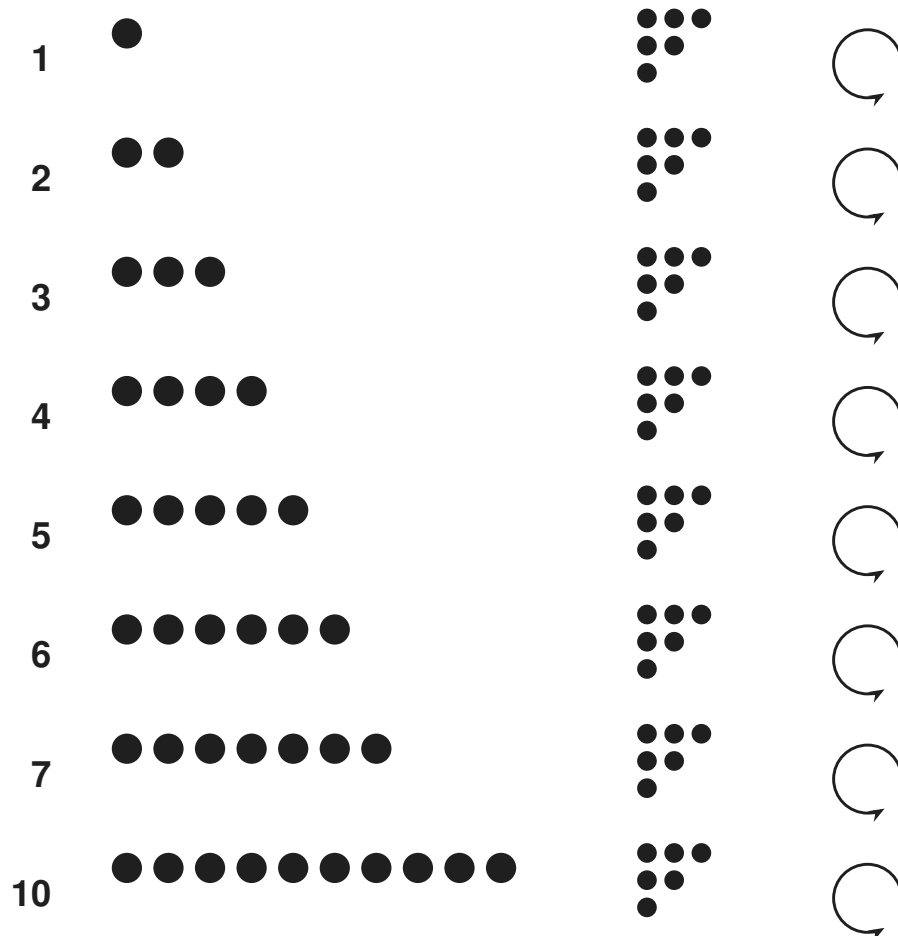
Problem 8: Put 5 counters in one row on the table. Make a move. Put the new picture on mat 1. Then do what you did in Problem 5: copy the newest mat with new counters, make a move, and put the new picture on the next mat. Stop when all 6 mats are full. Draw a line between two mats that show the same picture. Find every pair.

1	2
3	4
5	6

Problem 9: Put 2 counters in one row. Make a move and color box 1. Make a move and color box 2. Make a move and color box 3. Now put just 1 counter on the table. Make a move. Color box 4 to show what you get.



Problem 10: Each row below is a number of counters you have played with. If the game with that many counters stops changing, circle the stairs. If it goes round and round, circle the arrow. Fill the rows for 7 and 10 when you have played them.



Problem 11: Put 7 counters on the 7 big circles. Make a move and color box 1. Keep going. Color the next box after each move, up to box 9. Find two boxes that show the same picture and draw a line between them. Then look at boxes 3, 4, 5 and 6. In each one, draw a line around a stairs with rows of 3, 2 and 1, and put a ring around the circle that is left over.

